

Multiple Choice

1. **Answer:** A

Options: A) v^2 ; B) $v/4$; C) $v/2$; D) $4v$; E) $4v^2$

Note: Correct option: A - v^2

2. **Answer:** D

Options: A) 1.2×10^4 Joules; B) 3.6×10^4 Joules; C) 2.2×10^4 Joules; D) 1.72×10^4 Joules; E) 1.9×10^4 Joules

Note: Correct option: D - 1.72×10^4 Joules

3. **Answer:** A

Options: A) remains the same; B) Not enough information to say; C) is not affected; D) triples; E) decreases

Note: Correct option: A - remains the same

4. **Answer:** B

Options: A) Both of the above; B) +3 e and -4 e; C) +5 e and -6 e; D) +2.5 e and -3.5 e; E) +4 e and -5 e

Note: Correct option: B - +3 e and -4 e

5. **Answer:** D

Options: A) Breaking the speed of light; B) Flying slower than sound; C) pulling out of a subsonic dive; D) flying faster than sound; E) Flying at constant speed

Note: Correct option: D - flying faster than sound

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'speed'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '9 lb'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '3.23 km/s'.

Long Form Response

11. **Answer:** increased by a factor of 16. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 12.0 amp. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key